ANALYTICAL CUMULATIVE EXAMINATION

February 17, 1990

Briefly answer the following questions.

1. (20) Why will a nonlinear calibration curve and a loss in sensitivity occur in atomic absorption (AA) spectroscopy when using a continuum light source as opposed to a sharp line source?

2. (20) A 12 ppm solution of lead gives an AA signal of 30% absorption. A blank gives no signal. Calculate the AA sensitivity.

3. (20) Explain the following observation. A sodium solution is analyzed by flame emission using the 589 nm sodium doublet. A 1 ppm standard of sodium gives an emission reading of 112, whereas the same solution containing 10 ppm potassium gave a reading of 123. A 10 ppm potassium standard gave no reading at 589 nm.

4. (40) The 228.8 nm cadmium line corresponds to a $^1S_0$ -- $^1S_1$ transition. The ratio of the statistical weights of the upper to the lower state is 3. Calculate the ratio of the number of atoms in the upper to the lower state for an air acetylene flame (2250 ^oC).

$$N_u/N_o = (g_u/g_o)e^{-(E_u-E_o)/kT}$$

(c, 2.998 x 10^{10} cm/sec; k, 1.381 x 10^{-23} JK^{-1}; h, 6.626 x 10^{-34} Js)